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MAE-Optimized Machine Learning Ensemble for Post-Processing GFS Solar Irradiance Forecasts in East Asia

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Calibrate on the absolute error, not the squared error

In contrast to a typical calibration, the ensemble is trained on the absolute error instead of the squared error. Across three climates in East Asia, it reduces the error of raw GFS by up to 21%, with the lowest nMAE at every station.

11–18% lower nMAE than raw GFS

13–21% lower nRMSE than raw GFS

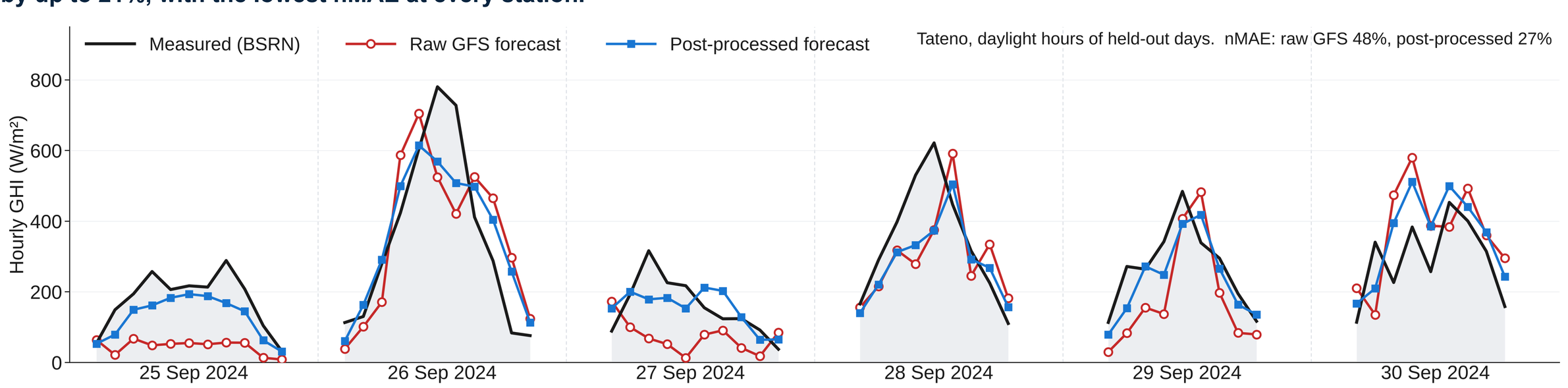
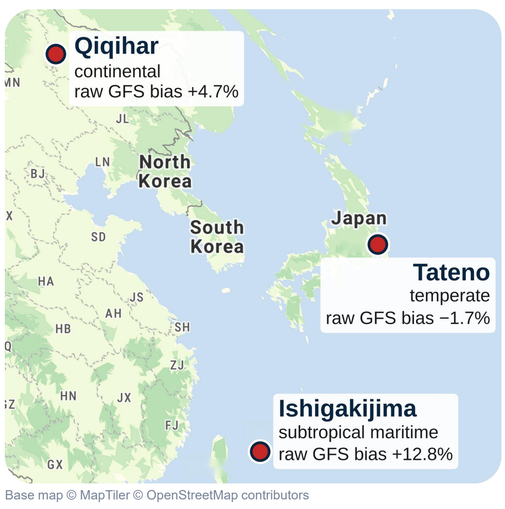


Fig. 1. On 25 and 27 September GFS forecast much thicker cloud than was observed; the calibration removes most of that error. Held-out days at Tateno, chosen for illustration; scores for the whole year are in Fig. 2.

Context

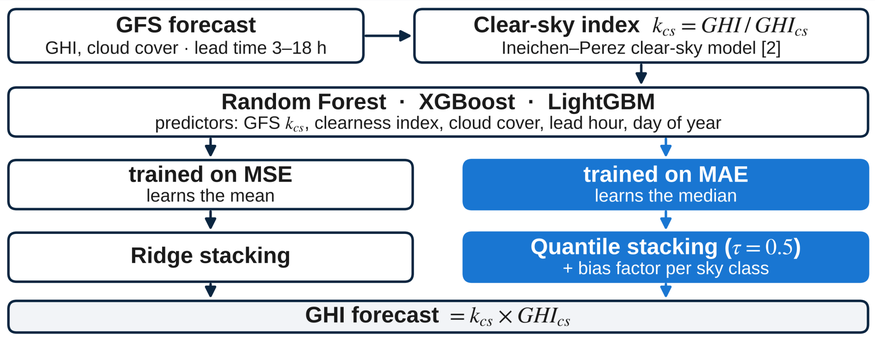


- GFS forecasts are free and global, but biased at all three stations (map).
- Machine-learning calibration against ground data reduces these errors [4].
- The loss function decides what the model learns: the squared error leads to the mean, the absolute error to the median [1].

Objective

Is an ensemble trained on the absolute error, from the base learners to the stacking, a better calibration of GFS than the traditional MSE-trained one?

Methodology



Data: BSRN 1-min GHI [3], 2024, hourly, Sun above 5°. **Reference:** MOS-MLR.
Test: 5-fold cross-validation in 6-day blocks. A random split of hours would have shown nRMSE 4–6 points lower, through leakage between neighboring hours.

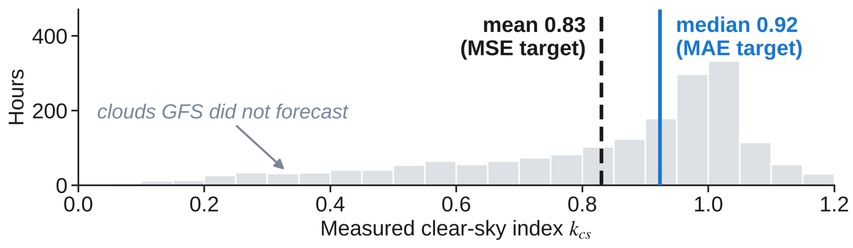


Fig. 3. When GFS forecasts a clear sky, one hour in four is in fact cloudy. These hours (k_{cs} < 0.7) split mean and median. Ishigakijima, $k_{cs,GFS}$ > 0.9, k_{cs} ≤ 1.2.

Results and Discussion

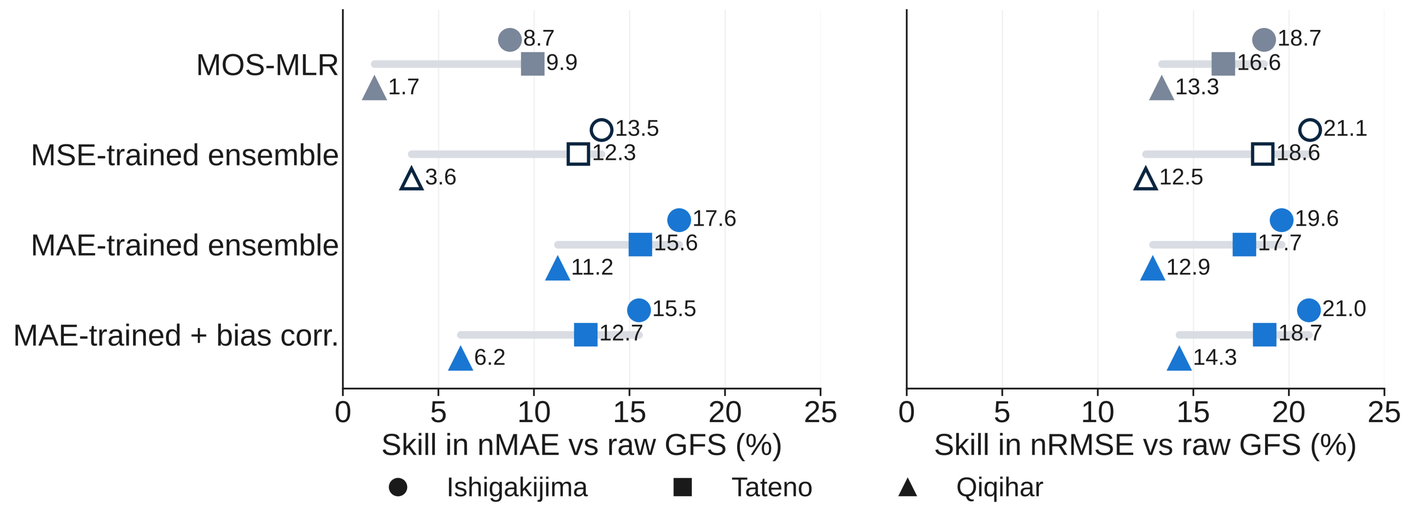


Fig. 2. MAE training gives the highest nMAE skill at every station; in nRMSE the ensembles are practically equivalent. Skill against raw GFS, all 2024 daylight hours, 5-fold blocked test.

nMAE (%)
nRMSE (%) below LightGBM (MAE)

Trained at \ Tested at	Ishigakijima	Tateno	Qiqihar
Ishigakijima	**25.8** 36.1	26.4 37.0	25.8 39.1
Tateno	29.4 43.4	**20.6** 32.4	19.3 31.5
Qiqihar	30.6 46.7	23.7 36.7	**19.5** 33.7
all three, pooled	28.2 41.9	20.9 33.3	18.9 31.4
pooled + station ID	**26.5** 38.4	**20.9** 33.6	**18.8** 31.9
raw GFS	**32.7** 48.8	**26.9** 43.3	**20.5** 36.0

Fig. 4. The calibration is site-specific: trained in the subtropics, it is worse than raw GFS at Qiqihar. A pooled model with a station identifier matches the local ones.

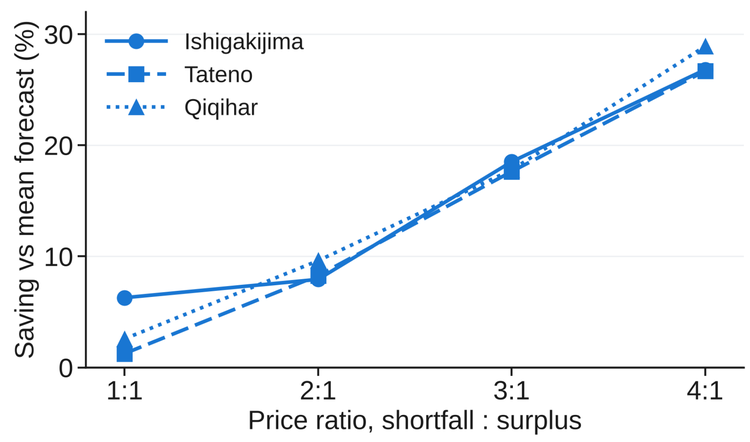


Fig. 5. The more asymmetric the imbalance price, the more the quantile forecast saves.
Quantile τ = 1/(1+r) instead of the mean, for a shortfall priced r times a surplus.

Conclusions

1. Calibration in clear-sky index space reduces the error of raw GFS at all three stations: nMAE by 11–18%, nRMSE by 13–21%.
2. Training on the absolute error lowers nMAE at every station; with a sky-class bias correction, nRMSE and bias also match the MSE-trained ensemble.
3. One pooled model with a station identifier is as accurate as the local models. With asymmetric imbalance prices, the matching quantile cuts the cost by 8–29%.

Limits: one year of data, three stations, lead time 3–18 h.

Acknowledgment

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